Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
roll_window: 1
timesteps_orig: 60
timesteps: 60
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
forecast_steps: 20

LSTM Multi-Step Training Report

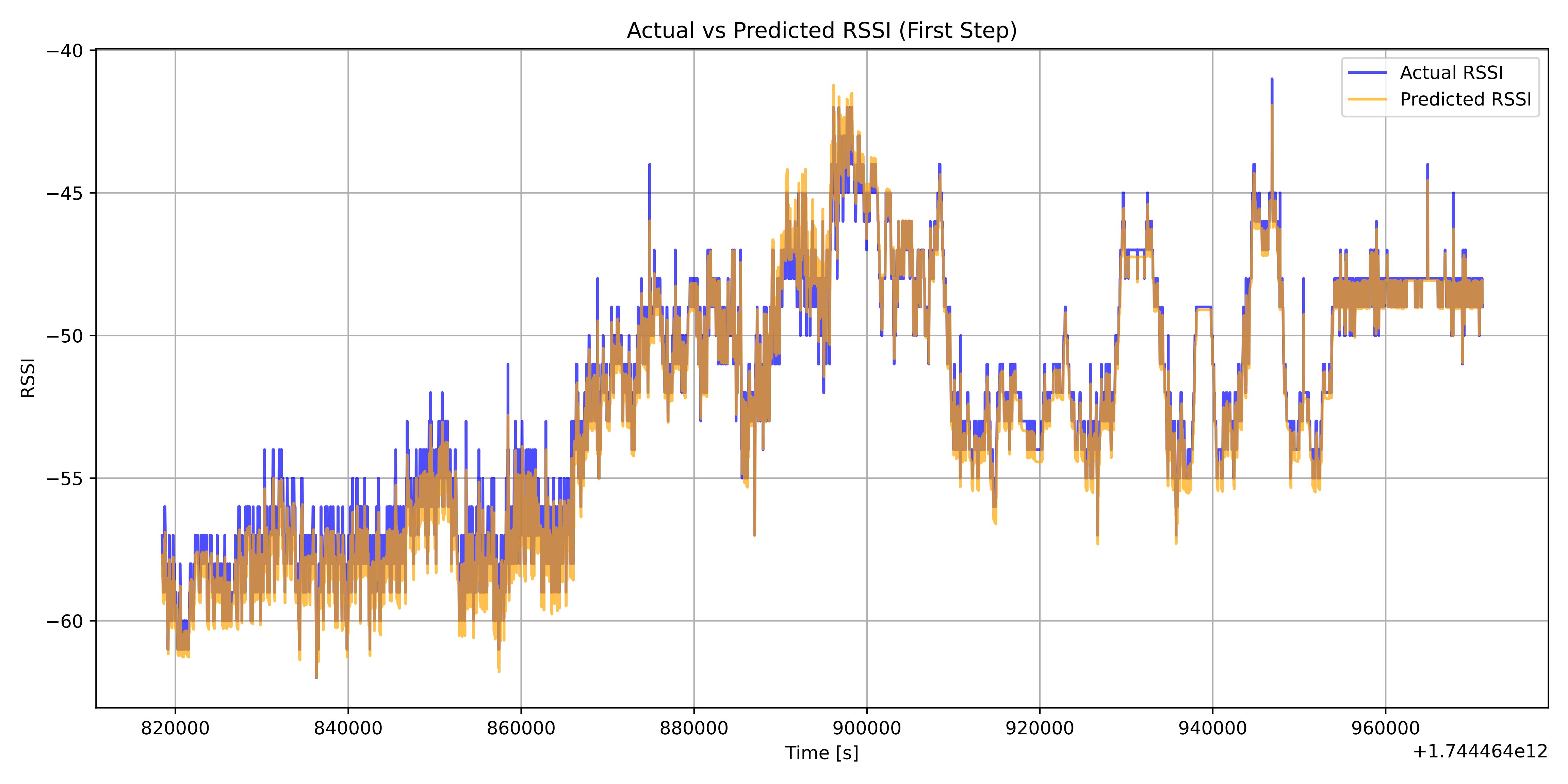
Report Generated: 2025-05-08 18:04:22

Training Start: 2025-05-08_16-35-49
Training End: 2025-05-08_17-38-46
Training Duration: 62.95 minutes

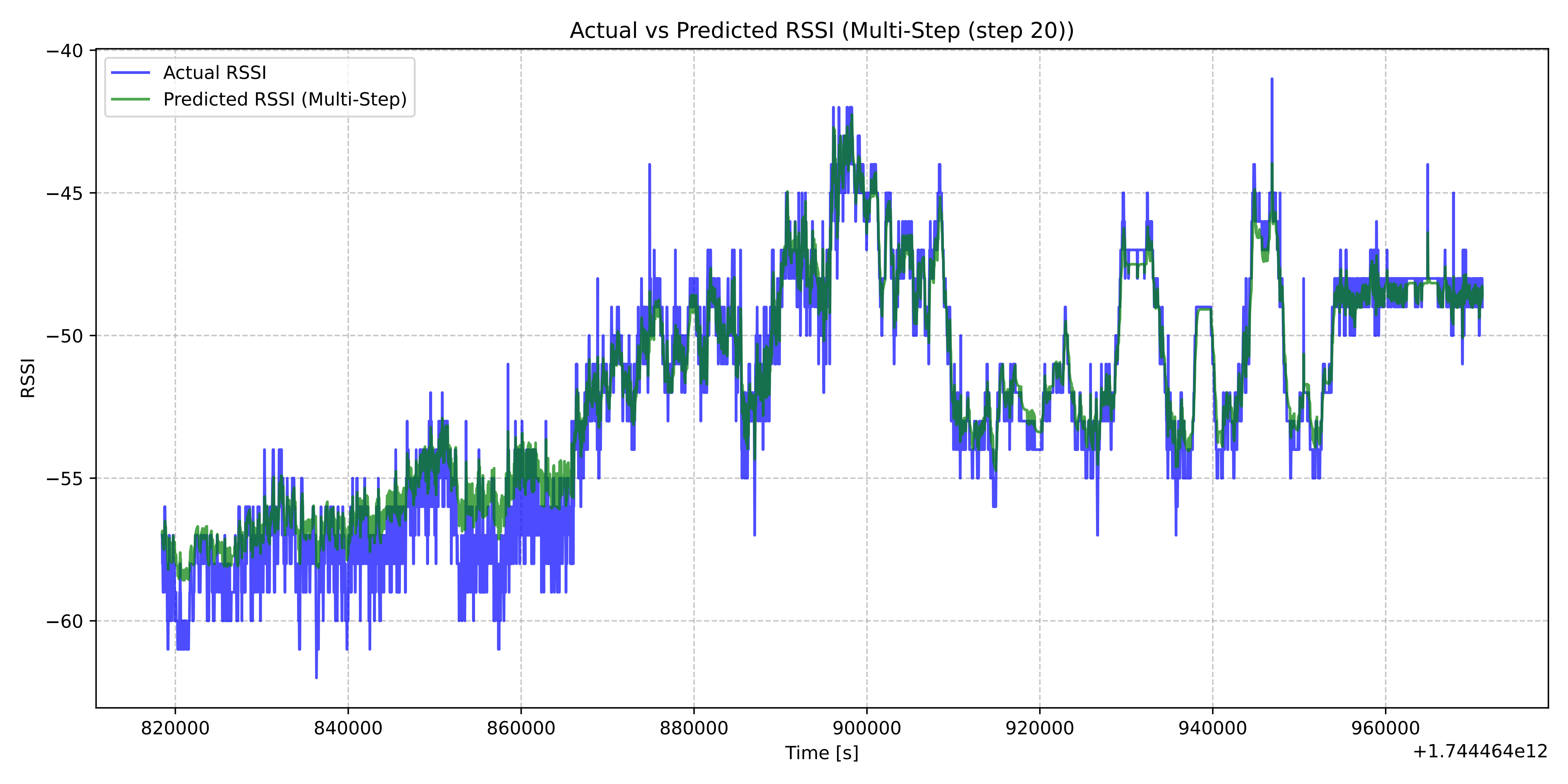
Best Hyperparameters:
Number of Layers: 1
Activation: tanh
Optimizer: Nadam
Dropout Rate: 0.20

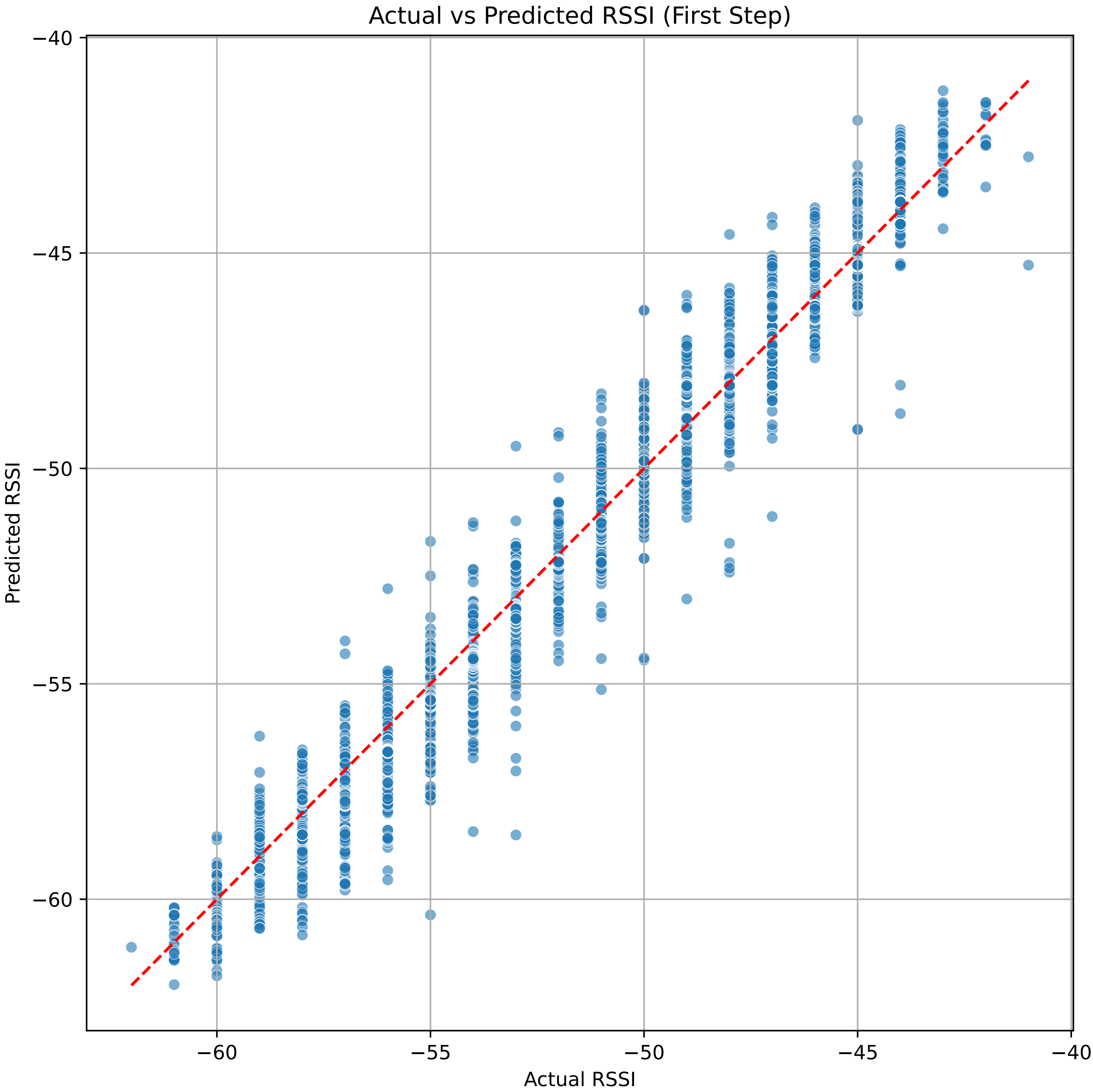
MAE: 0.8910
MSE: 1.4070
RMSE: 1.1862
R2 Score: 0.9222

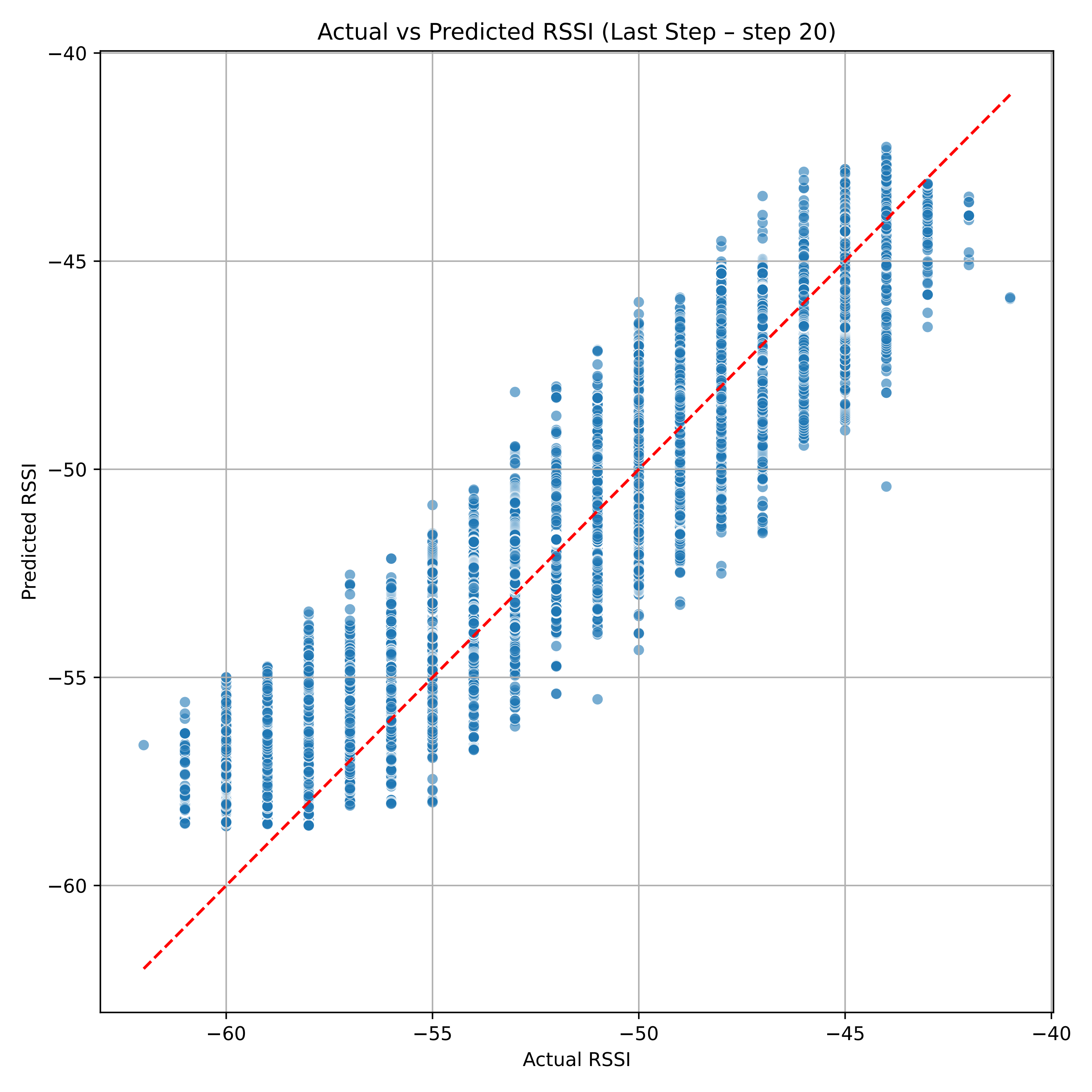
Multi-Step Forecast:
Forecast Steps: 20
MAE: 0.0079
MSE: 0.0001
RMSE: 0.0107
R2 Score: 0.9718



Actual vs Predicted RSSI (First Step)
Actual RSSI
Predicted RSSI
RSSI
−40
−45
−50
−55
−60
820000
840000
860000
880000
900000
920000
940000
960000
Time [s]
+1.744464e12

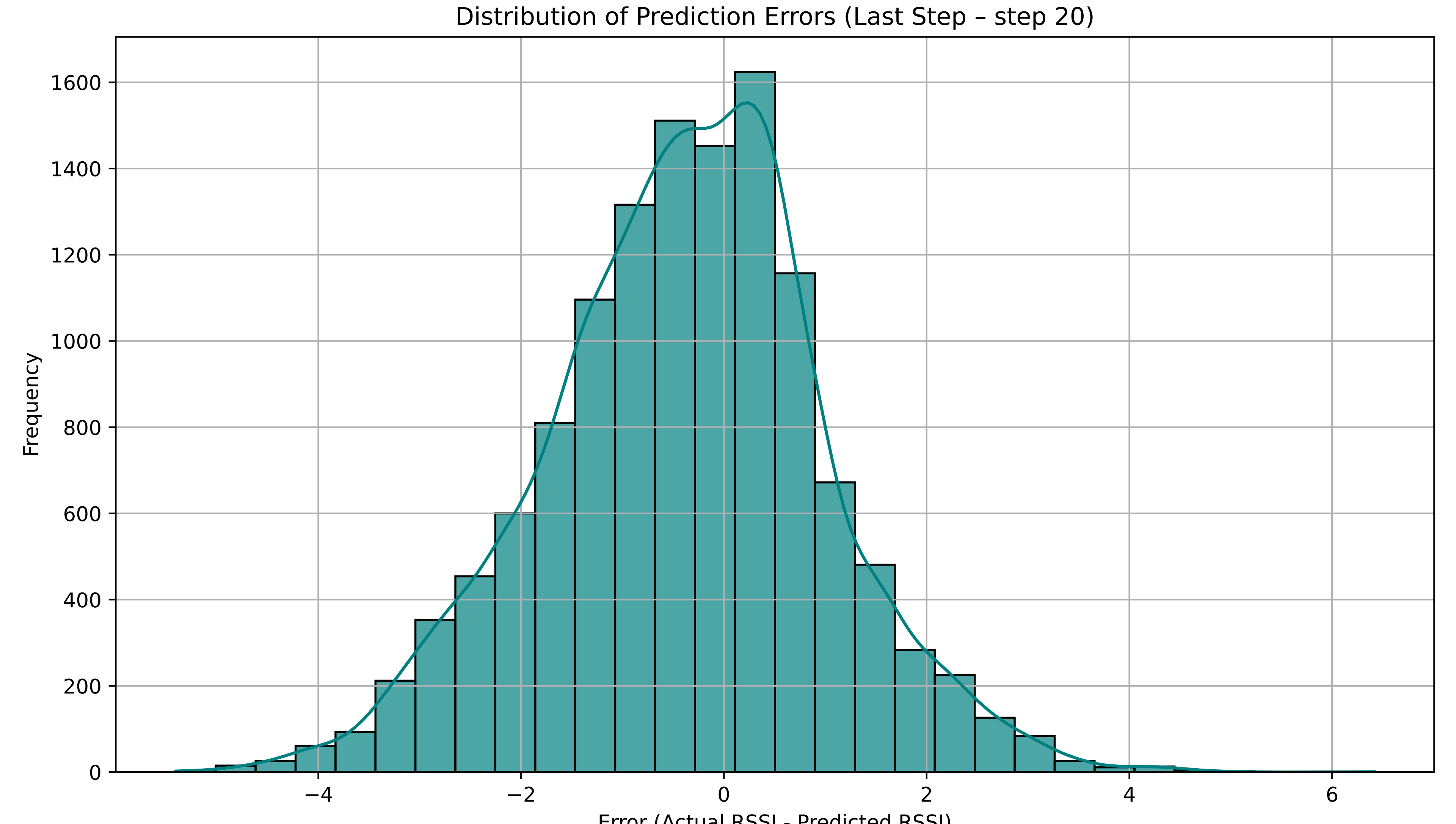

Actual vs Predicted RSSI (Multi-Step (step 20))
Actual RSSI
Predicted RSSI (Multi-Step)
−40
−45
−50
−55
−60
RSSI
820000
840000
860000
880000
900000
920000
940000
960000
Time [s]
+1.744464e12


Actual vs Predicted RSSI (First Step)
Predicted RSSI
Actual RSSI
−40
−45
−50
−55
−60
−60
−55
−50
−45
−40

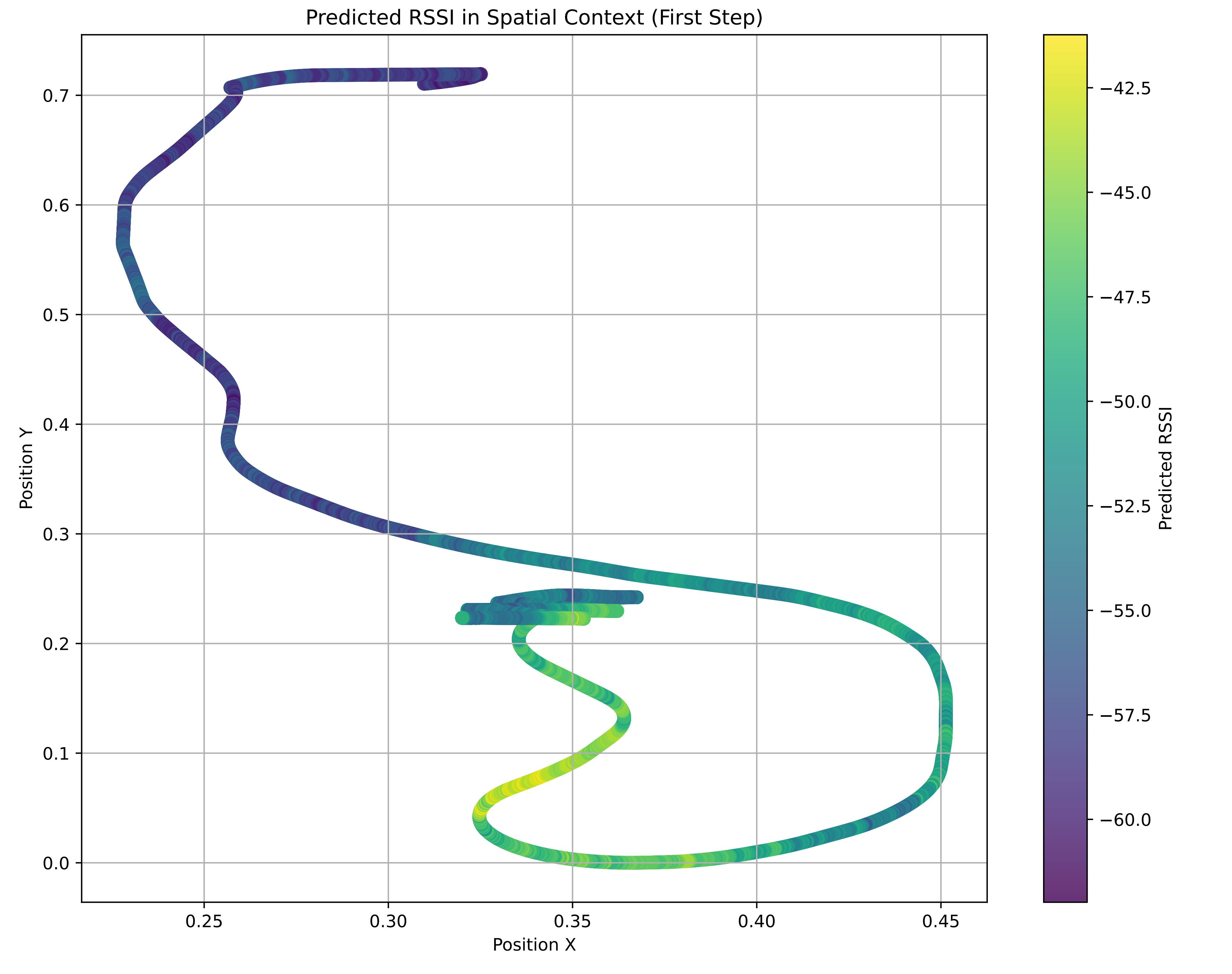


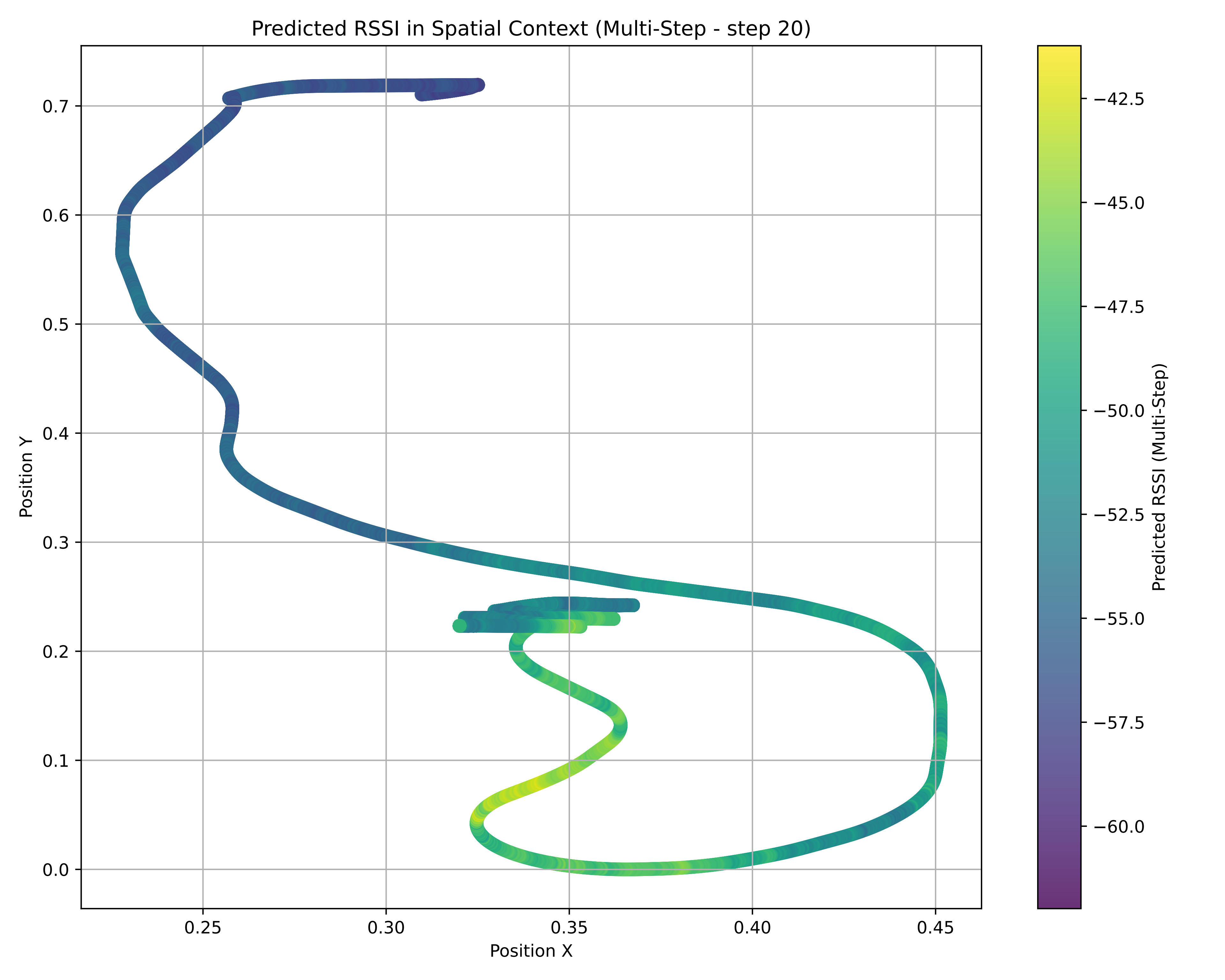
Actual vs Predicted RSSI (Last Step – step 20)
Predicted RSSI
Actual RSSI
−40
−45
−50
−55
−60
−60
−55
−50
−45
−40

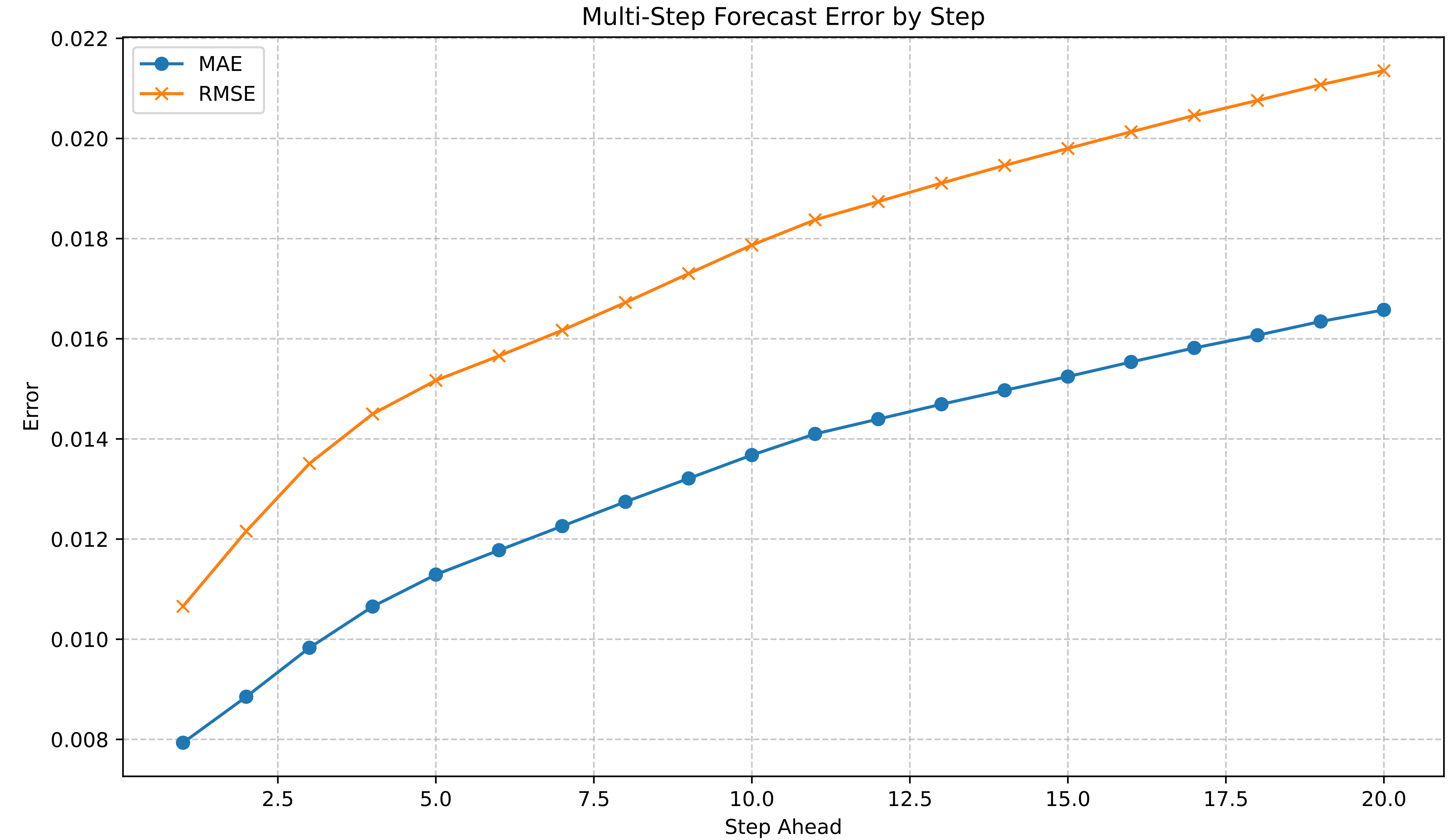

Distribution of Prediction Errors (First Step)
Frequency
4000
3500
3000
2500
2000
1500
1000
500
0
−4
−2
0
2
4
Error (Actual RSSI - Predicted RSSI)



Distribution of Prediction Errors (Last Step – step 20)
Frequency
1600
1400
1200
1000
800
600
400
200
0
−4
−2
0
2
4
6
Error (Actual RSSI - Predicted RSSI)


Predicted RSSI in Spatial Context (First Step)
Position Y
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Position X
0.25
0.30
0.35
0.40
0.45
Predicted RSSI
−42.5
−45.0
−47.5
−50.0
−52.5
−55.0
−57.5
−60.0


Predicted RSSI in Spatial Context (Multi-Step - step 20)
Position Y
Position X
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.25
0.30
0.35
0.40
0.45
Predicted RSSI (Multi-Step)
−42.5
−45.0
−47.5
−50.0
−52.5
−55.0
−57.5
−60.0



Multi-Step Forecast Error by Step
MAE
RMSE
Error
Step Ahead
0.022
0.020
0.018
0.016
0.014
0.012
0.010
0.008
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

```
Resumo Automático
-----------------
Dataset:        rssi_odom_dataset_12apr25_ajust.parquet
Modelo:         LSTM
Forecast steps: 20

MAE / RMSE (dB)
    • 1º passo : 0.01 / 0.01
    • Mediana  : 0.01 / 0.02
    • 20º passo : 0.02 / 0.02

Crescimento do erro: 109.0%   →   Degrada Rápido (> 50 %)

3 piores passos (MAE): 20 (0.02), 19 (0.02), 18 (0.02)

Recomendação:
Considere aumentar o histórico de entrada ou usar um modelo mais
profundo para melhorar multi-step.
```